

PROFILE

Production AI and platform engineer with about eight years of experience delivering enterprise and user-facing systems across application software, cloud platforms, security, and operations.

Carries work from architecture and implementation through integration, rollout, observability, incident response, and production stabilization.

SELECTED EVIDENCE

AWS EKS platform for governed, self-service AX application delivery	Production reliability for an enterprise AI service used by 30,000 employees	Linux packet application measured at up to 130,000 packets per second
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CORE SKILLS

Cloud & Platform	AWS EKS, Azure, Kubernetes, OpenShift, Docker, Helm/Helmfile, ArgoCD, GitOps
Software	Python, Bash, Linux, Kubernetes Operators, API and runtime integration, asynchronous flows
Reliability & Security	Grafana, Prometheus, OpenTelemetry, incident response, authentication, authorization, security governance
Applied AI Systems	Agent application delivery, LLM evaluation with MLflow, existing LLM Gateway API integration, model inference

EXPERIENCE

Samsung Electronics, AX Development Group	Seoul, South Korea
SRE / AI Platform Engineer	Jan. 2026 - Present
- Design and build an internal AX Agent Platform on AWS EKS, adapting the agent runtime and deployment path to enterprise platform standards. - Implemented self-service AX application delivery with automatic domain issuance, authorized-only access, and platform controls for authentication, authorization, security, and governance. - Enabled AX applications to call the existing enterprise LLM Gateway API through platform-side client compatibility and runtime integration. - Supported a Microsoft-collaborative B2C generative AI service through UK rollout readiness, initial production operations, incident response, and stabilization from Jan. to Jun. 2026.	
Samsung Research	Seoul, South Korea
SRE / AI Platform Engineer	Jul. 2024 - Dec. 2025
- Led production delivery and application reliability for a company-wide AI chatbot serving 30,000 employees on internal Kubernetes clusters. - Owned releases across development, staging, and production while integrating application delivery with security and operational requirements. - Built container delivery pipelines with Docker, Kubernetes, GitHub Actions, ArgoCD, Helm, and Helmfile. - Deployed Grafana, Prometheus, and OpenTelemetry observability and built a Python/MLflow evaluation workflow for LLM-based machine translation.	
Samsung Electronics, Network Division	Seoul, South Korea
SRE & Software Engineer	Jan. 2021 - Jun. 2024
- Designed and delivered containerized applications with Docker, Helm, Kubernetes Operators, and AWS services including EKS, ECS, Route 53, RDS, and IAM. - Built and managed OpenShift, Kubernetes, and EKS environments for reliability, high availability, performance, and auto-scaling. - Designed and developed a Bash-based one-click Kubernetes cluster provisioning system for cloud developers. - Developed a Linux application that mirrors and forwards up to 130,000 packets per second under constrained system resources.	

EXPERIENCE (CONTINUED)

TmaxOS & TmaxCloud

Gyeonggi, South Korea

Software & Platform Engineer

Jan. 2019 - Dec. 2020

- Built a Siri-like AI assistant on TmaxOS, covering model inference, data preprocessing, system integration, and asynchronous application flows.
- Designed and developed Kubernetes CI/CD and registry operators and deployed a private container image registry.

SELECTED ENGINEERING WORK

Enterprise AI Assistant | Public reference implementation

jinwonlee.github.io/projects/enterprise-ai-assistant/web/

Built an access-aware Python assistant with SQLite FTS5 retrieval, pre-generation role filtering, grounded citations, audit events, evaluation cases, a FastAPI surface, and an optional OpenAI Responses API provider.

LLM Performance Evaluation Pipeline

Built a Python and MLflow workflow for repeatable machine-translation model evaluation and result comparison on Kubernetes.

Distributed CNN Training Research

Contributed to resource-aware layer placement for distributed TensorFlow training, reporting up to 2.3x training speedup in published research artifacts.

High-throughput Packet Mirroring

Developed and tuned a Linux data-path application that mirrors and forwards up to 130,000 packets per second under constrained resources.

EDUCATION & RESEARCH

UNIST | M.S. in Computer Science and Engineering

Mar. 2017 - Feb. 2019

Master's thesis: Convergence Aware CNN Training | Advisors: Jiwon Seo and Sam H. Noh

UNIST | B.S. in Computer Science and Engineering, Minor in Mathematics

Mar. 2012 - Feb. 2017

Undergraduate research: distributed systems for AI model training with TensorFlow

PUBLICATIONS & LANGUAGE

- USENIX NSDI 2019 poster: Alleviating the Network Bottleneck for CNN Distributed Training through Automatic Resource-Aware Layer Placement.
- arXiv 2019: Accelerated Training for CNN Distributed Deep Learning through Automatic Resource-Aware Layer Placement.
- ACM EuroSys 2018 poster: Improving Performance of Distributed TensorFlow using CNN Characteristics Exploiting Model Parallelism.
- OPlc IH (Intermediate High), Dec. 2025.